ANALYSIS REPORT

Wave 5.2R Stage 1 Extended Scientific Technique Discovery

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Decision

Stage 1 is complete and passes its exit gate.

The search did not identify a previously missed, complete governing equation that can be applied one-to-one to the current test-rig dataset. It did identify a broad, scientifically grounded set of ways to help a neural model without pretending that unavailable contact or internal-state variables are observed.

The candidate register is therefore frozen with:

- `30` documented techniques;
- `13` techniques in the active real-data roster;
- `11` techniques retained behind explicit diagnostic gates;
- `3` synthetic or offline oracle techniques;
- `3` excluded techniques;
- all `13` roadmap search families covered;
- zero target-derived runtime variables;
- zero missing variables in the active or conditional real-data roster.

  The decision is not that physics is useless. The decision is that different forms of knowledge must be used at different strengths:

1. exact construction for known periodic structure;
2. supervised spectral and derivative guidance for directly measurable curve

   properties;
3. bounded discrepancy learning around the Polynomial-Fourier anchor;
4. weak, falsifiable priors for incomplete mechanics;
5. synthetic or offline tests for mechanisms that are not identifiable from

   the current measurements. This hierarchy lets the network compensate for an incomplete analytical model while preventing the neural residual from hiding an invalid physical claim.

Scope And Frozen Contract

Stage 1 inherits the immutable Stage 0 contract.

Contract item	Frozen value
Dataset	`polished_dataset`
Input mode	setpoints
Surface	`Fw` only
Split signature	`c1aa8718fb9bf88cc2021c121dc4f3b4010fc1d2e45ac90af5f4376aa64f8e16`

Contract item	Frozen value
Test conditions	`97`
Runtime variables	angle, speed setpoint, torque setpoint, temperature setpoint
Analytical anchor	`PF_A_LOCAL_QUADRATIC`
Neural references	accepted periodic harmonic MLP and accepted periodic GRU

Training curves may supply measured TE, harmonic coefficients, angular derivatives, PF-A residuals, and diagnostic statistics. Those quantities may define training targets or losses, but they are not allowed as runtime inputs.

The following states are not available causally:

- contact force and contact location;
- mesh stiffness and internal load sharing;
- component-level manufacturing error groups;
- friction state and internal loss power;
- causal acceleration and identified inertia;
- repeated reversal and loop history;
- synchronized electrical current and motor sidebands;
- wear chronology and unit-specific geometry.

  Any formulation requiring those quantities is excluded from real-data training or retained only as a synthetic oracle.

Evidence Method

The review combined three evidence layers.

Repository-Primary Evidence

The ingested Bauer Polynomial-Fourier material, Fourier and polynomial notes, the recovered Matlab implementations, RV reducer mechanics summaries, MMT mapping, hysteresis and backlash synthesis, and dataset contract were read as project-specific evidence.

Stage 0 was treated as the empirical baseline. It establishes that:

- the analytical PF-A anchor is reproducible;
- the accepted GRU is strongest on most raw and shape metrics;
- the harmonic MLP remains strong in harmonic diagnostics;
- no later candidate may claim improvement by changing the split or angular

  convention.

External Primary Literature

Primary papers and official proceedings were searched for:

- model discrepancy and grey-box learning;
- variational and weak-form PINNs;
- loss balancing and gradient conflict;
- spectral bias and coordinate networks;
- derivative-aware objectives;
- symbolic and sparse discovery;
- uncertainty-aware weighting;
- curriculum and adaptive sampling;
- certification and neural operators;
- hybrid mechanical-system identification.

Review articles and secondary summaries were not used as the scientific basis for candidate inclusion.

Translation Rule

A paper result was not copied directly into the local plan. Each method was translated through four questions:

1. Does the method address a failure that can exist in this TE task?
2. Are all required variables available causally or derivable from training data only?
3. Can the local hypothesis be compared with a matched control?
4. Is there a result that would falsify the local hypothesis?

Only techniques answering all four questions enter the real-data roster.

What Counts As Physics Guidance Here

The word physics covers several distinct kinds of structure. Keeping them separate is essential.

Exact Observable Structure

The angular curve is periodic and has a measurable harmonic decomposition. These are directly testable properties of the output. They can be imposed through:

- periodic features;
- exact periodic output construction;
- complex harmonic targets;
- spectral losses;
- zero-mean centered-shape construction.

These constraints are strong because they do not require unobserved internal states.

Semi-Analytical Mechanistic Structure

PF-A predicts Fourier coefficients as low-order functions of operating conditions. It is an effective analytical anchor, but it is not a complete governing law for every reducer mechanism.

The appropriate hybrid is `y_hat(theta, z) = PF_A(theta, z) + bounded_residual(theta, z)` .

where `z` contains the three setpoints. The residual is allowed to repair misspecification, but it must beat a parameter-matched data-only control and remain bounded under anchor corruption.

Weak Mechanical Priors

Quasi-static compliance, torque sensitivity, smooth coefficient surfaces, and bounded response are physically motivated but incompletely specified. They may be tested as weak sign or interval priors only after training-data bootstrap analysis supports them.

They are hypotheses, not laws. If the weak prior reduces its own loss without improving held-out curves, it is rejected.

Unobservable Governing Mechanisms

Contact, load sharing, friction energy, dynamic inertia, MMT component errors, and backlash memory are valid mechanical topics. They cannot be evaluated honestly from the current forward setpoint contract.

The network cannot make a missing state observable merely because a loss equation contains that state. Replacing the missing state with target TE would create leakage and circular reasoning.

Frozen Technique Roster

Active Real-Data Roster

These techniques are directly testable with the frozen contract.

ID	Technique	Priority	Stage	Main test
T01	Bounded PF-A discrepancy residual	high	3	Anchor plus residual versus matched residual control
T02	Complex harmonic-coefficient residual	high	5	Joint amplitude and phase fidelity
T03	Mean and centered-shape heads	high	7	Offset and shape improve jointly
T04	Fourier-feature coordinate residual	high	4	High-order fidelity without spectral artifacts
T05	SIREN coordinate residual	medium	4	Derivative and harmonic gain versus tanh
T07	Complex spectral supervision	high	6	Harmonic gain without raw-error regression
T08	Angular Sobolev supervision	high	6	Derivative gain robust to estimator choice
T09	Local Fourier-moment weak residual	medium	13	Integrated moments improve held-out curves
T11	Gradient-statistics annealing	high	2	Reduced loss-gradient imbalance
T12	ReLoBRaLo balancing	high	2	Stable weights and better repeatability
T16	Staged guidance curriculum	high	2	Better final result at equal update budget
T22	Temporal PF-A plus GRU residual	high	9	Beat the accepted GRU, not only PF-A
T23	Exact periodic output construction	high	4	Closure by construction without lost capacity

Conditional Real-Data Roster

These techniques remain testable, but only after their trigger is observed.

ID	Technique	Trigger	Priority
T06	Wavelet coordinate network	Stable localized multiscale residual remains after Fourier and SIREN tests	low
T10	Weak compliance sign or bound	Training bootstrap establishes a stable derivative sign or interval	medium
T13	Main-loss-preserving PCGrad	Persistent negative gradient cosine is measured	medium
T14	CAGrad	Multi-objective conflict remains after simpler adapters	low
T15	Augmented Lagrangian	A simple soft prior first shows incremental signal	low
T17	Failure-informed sampling	Cross-fitted localized failure regions are stable	medium
T18	Self-adaptive point weights	Effective-sample-size and noise checks prevent collapse	low
T19	Ensemble anchor-trust calibration	Error localization is useful enough to justify inference cost	medium
T20	Sparse harmonic-condition discovery	Terms are stable across bootstrap resamples	medium
T21	Constrained symbolic residual search	Sparse-library limits are demonstrated	low
T24	Residual certification	A supported tournament candidate and meaningful residual exist	low

Oracle-Only And Excluded Roster

ID	Technique	Decision	Reason
T27	Misspecified-anchor synthetic curriculum	oracle only	Useful for recovery power, not proof of real mechanics
T28	Paper-faithful MMT oracle	oracle only	Component errors and geometry are not causally observed
T29	Contact, friction, and energy residuals	oracle only	Internal force, stiffness, sharing, and loss states are absent
T25	Physics-informed neural operator	excluded	No function-valued excitation or governing PDE operator
T26	Universal differential-equation residual	excluded	No identified dynamic state, acceleration, or initial-state contract
T30	Hysteresis and backlash memory	excluded	Forward-only curves do not contain repeated reversal loops or reset state

Detailed Formulations

Analytical Discrepancy Learning

Model-discrepancy PINNs explicitly distinguish an assumed model from a learned correction. That is the closest scientific analogue to the local problem: PF-A explains a large part of the signal, while a network represents what the polynomial coefficient surface omits.

The primary candidate is

- inputs:

  `z = [speed_setpoint, torque_setpoint, temperature_setpoint]` ;
- hybrid:

  `y_hat = PF_A(theta, z) + b(z) tanh(r(theta, z) / b(z))` . The envelope `b(z)` must be estimated from training residuals only. The unbounded version is retained as an ablation, not as the default. Required controls are:
- PF-A alone;
- a parameter-matched data-only residual network;
- the same hybrid with a corrupted or omitted anchor;
- the accepted harmonic MLP and accepted GRU.

  The hybrid is falsified if additional capacity, rather than the analytical anchor, explains the gain. It is also falsified if the residual becomes an unbounded replacement for PF-A.

Complex Harmonic Coefficients

For harmonic order `k` , represent the coefficient as `C_k = A_k exp(j phi_k) = Re(C_k) + j Im(C_k)` .

Optimize `L_complex = sum_k w_k |C_hat_k - C_k|^2` .

This avoids the discontinuity between phases near `-pi` and `pi` . The weights are fitted or normalized on training data and frozen before held-out evaluation.

Two related candidates are kept separate

- `T02` predicts corrections to complex coefficients directly;
- `T07` applies a training-only complex spectral loss to a curve predictor.

  The separation determines whether any gain comes from the representation or from the objective.

Mean And Centered Shape

The multi-head construction is

- centered shape:

  `s_centered(theta,z) = s_raw(theta,z) - mean_theta(s_raw(theta,z))` ;
- reconstructed curve:

  `y_hat(theta,z) = mu_hat(z) + s_centered(theta,z)` . The zero-mean constraint is architectural and exact on the sampled angular grid. It prevents the shape head from silently carrying the offset. The matched single-head control must have comparable parameter count. A result is accepted only when offset and centered-shape metrics improve jointly.

Fourier Features And Exact Periodicity

For a predeclared set of angular orders `K`, the coordinate map is `gamma(theta) = [sin(k theta), cos(k theta) for k in K]`.

This directly exposes the periodic basis and guarantees identical features at `theta` and `theta + 2*pi`. It is preferable to learned arbitrary frequencies as the first test because the reducer harmonics are already measurable.

The experiment must compare

- fixed physically motivated orders;
- a raw-angle MLP;
- learned or random Fourier frequencies at the same parameter budget;
- exact periodic construction versus a soft closure penalty.

SIREN And Wavelets

SIREN is retained because periodic activations can represent fine detail and derivatives. Its initialization is part of the method and cannot be replaced with an arbitrary sinusoidal MLP initialization.

Wavelets are lower priority. The current TE signal already has a strong global harmonic representation. A wavelet network is justified only if Stage 2 and Stage 6 show stable, localized multiscale residuals that Fourier features and SIREN cannot resolve.

Sobolev Guidance

The first-order angular objective is `L_d1 = mean((d y_hat / d theta - d y_train / d theta)^2)`.

A second-order term is allowed only after:

- the derivative estimator is selected on training data;
- smoothing sensitivity is reported;
- gains reproduce under at least two reasonable derivative estimators;
- the model does not amplify unsupported high-frequency noise.

  Derivative loss is training-only. Runtime inference still requires only angle and setpoints.

Weak-Form Harmonic Moments

For residual `r(theta,z)`, fixed test functions `psi_m(theta)` define:

- weak moment: `R_m(z) = integral r(theta,z) psi_m(theta) d theta`;
- weak loss: `L_weak = sum_m |R_m(z)|^2`.

  The initial test set uses sine, cosine, and compact angular functions. The pointwise PF-A residual is the matched control. A small weak residual is not sufficient evidence. The method advances only if the integrated constraint improves held-out curve metrics and remains robust to a second predeclared test-function bank.

Weak Compliance

The theoretical mechanics support torque-dependent deformation, but the current data do not identify internal contact stiffness.

The only admissible real-data test is therefore a weak prior on a measurable response derivative, such as `d mu_hat(z) / d torque`.

The sign or interval must first be stable across training-only bootstrap resamples. A shuffled-torque prior is required as a negative control. No parameter may be labelled physical stiffness without independent mechanical identification.

Temporal Analytical Residual

The strongest temporal candidate combines PF-A with a recurrent residual: `y_hat_1:N = PF_A_1:N + bounded_GRU_residual_1:N`.

The hidden state must have deterministic initialization and reset semantics. The candidate must beat:

- PF-A;
- the accepted GRU;
- a parameter-matched GRU without PF-A.

  Beating only the analytical anchor is not enough.

Optimization And Sampling Findings

Instrument Before Intervention

The literature consistently shows that composite objectives can fail because their gradients differ in scale or direction. Stage 2 must therefore record:

- each named loss;
- gradient norm per loss;
- pairwise gradient cosine;
- update-to-parameter ratios;
- adaptive weights and their effective range.

  No gradient method is authorized merely because it is fashionable.

Fixed And Adaptive Weighting

The mandatory ordering is

1. equal weights;
2. manually normalized fixed weights;
3. gradient-statistics annealing;
4. ReLoBRaLo relative-progress balancing;
5. conflict-aware methods only after conflict is measured.

   Every adaptive method must preserve a minimum contribution from the primary data objective. Non-finite weights or data-loss collapse are immediate failures.

Gradient Conflict

PCGrad is modified locally so the data gradient is never projected. Only a secondary guidance gradient may be removed from a direction that conflicts with the primary objective.

CAGrad is a later, heavier comparator. It is not part of the first instrumentation implementation because its additional complexity is unjustified until multi-objective conflict is observed.

Augmented Lagrangian

An augmented Lagrangian cannot make a wrong equation informative. It may be used only after a weak or exact constraint shows incremental held-out signal under a simple soft formulation.

Improving constraint feasibility while prediction worsens is a failed result, not a successful PINN.

Curriculum

The retained schedule is

1. learn the data-only curve;
2. activate the analytical residual;
3. activate spectral or derivative guidance;
4. test freeze and unfreeze variants.

   The control receives the same total number of updates with all losses active from the start.

Adaptive Sampling

Failure-informed and retain-resample-release methods can be translated to:

- angular locations;
- operating-condition cells;
- harmonic-band failures.

  Scores must be cross-fitted on training data. Uniform-coverage samples remain in every batch so the sampler cannot erase easy regions and create blind spots.

Sparse And Symbolic Discovery

Sparse discovery is retained as an interpretation and candidate-generation tool, not as automatic discovery of a physical law.

The initial dictionary contains

- polynomial speed, torque, and temperature terms;
- selected cross terms;
- fixed harmonic orders;
- condition-by-harmonic interactions;
- PF-A coefficient residuals.

Terms must pass bootstrap stability selection. Stable empirical terms can become explicit Stage 10 ablations. Symbolic regression follows only after this restricted library establishes the limits of the simpler approach. Expressions must be:

- complexity bounded;
- unit safe;
- stable across resamples;
- evaluated on nested held-out conditions;
- compared with a parameter-matched MLP.

An interpretable expression is still empirical until independently connected to a reducer mechanism.

Uncertainty And Trust

Uncertainty is useful only if it locates error.

The retained tests are

- deep-ensemble spread;
- condition-distance uncertainty;
- PF-A discrepancy magnitude;
- calibration by speed, torque, and temperature band;
- high-error capture rate;
- uncertainty-error rank correlation.

A uniformly wide interval fails the gate. Ensemble deployment cost is high, so a full ensemble can advance only if its error localization is materially useful. A distilled trust head would then need its own calibration and parity test.

Certification

Post-training certification is retained as a late-stage option. It cannot certify that PF-A is the true governing physics. It may bound a selected network residual over a normalized input domain if:

- the architecture is supported by the verifier;
- the residual has a meaningful local interpretation;
- the bound is non-vacuous;
- empirical TE error is evaluated separately.

Dense adversarial gridding is the matched control.

Why Neural Operators Are Excluded

PINO and related neural operators learn mappings from functions to functions over families of parametric PDEs. The current task maps four finite-dimensional causal inputs to a periodic curve and does not expose:

- a function-valued excitation;
- a governing PDE;
- boundary-condition fields;
- multiple spatial resolutions of the same operator solution.

Using a neural operator here would add complexity without matching the local mathematical problem. It can be reconsidered only if the instrumentation and data contract change.

Why Full Dynamic, MMT, Contact, And Backlash Losses Stay Out

Universal Differential Equations

Universal differential equations require a partially known differential system and observable state trajectory. The polished forward setpoint lane does not identify acceleration, inertia, damping state, or initial-state semantics.

MMT

The MMT mapping remains scientifically valuable as a synthetic oracle. Paper-faithful real-data calibration is still deferred because the five equivalent-error groups and unit geometry are not independently observed.

Contact And Energy

Contact, load sharing, friction, and energy equations are valid. The dataset does not contain their internal forces, stiffnesses, losses, or geometry. A network output cannot be called those quantities without independent labels or an identifiable balance.

Backlash And Hysteresis

Backlash is direction and history dependent. Forward-only curves do not expose repeated reversals, minor loops, warm-up state, or deterministic reset semantics. A backlash loss would therefore encode an assumed hidden state that cannot be falsified in this lane.

Controls And Falsification Contract

Every later candidate inherits these requirements.

Mandatory Controls

- accepted GRU;
- accepted harmonic MLP;
- PF-A;
- parameter-matched data-only architecture;
- identical architecture without the proposed guidance;
- shuffled or corrupted prior when meaningful;
- equal total training budget.

Mandatory Metrics

- raw MAE and RMSE;
- absolute offset error;
- centered-shape error;
- peak-to-peak error;
- harmonic amplitude error;
- circular harmonic phase error;
- derivative and smoothness diagnostics;
- closure mismatch;
- robustness by operating-condition cell;
- seed stability;
- inference cost and deployment feasibility.

General Rejection Rules

A technique is rejected if

- its own loss improves but held-out curve metrics do not;
- it beats PF-A but not its matched neural control;
- it uses a target-derived runtime quantity;
- it depends on a missing state;
- gains disappear across fixed seeds;
- improvement comes from extra parameters or training steps;
- the inference graph cannot be made inspectable;
- the result requires post-hoc test-set tuning.

Deployment Assessment

The most deployment-friendly retained mechanisms are

- fixed Fourier features;
- complex coefficient heads;
- mean plus centered-shape decomposition;
- sparse explicit coefficient laws;
- bounded feedforward PF-A residuals.

Moderate-risk mechanisms are:

- SIREN activations, pending ONNX and Beckhoff support;
- PF-A plus feedforward residual composition.

High-risk mechanisms are:

- recurrent analytical residuals with reset state;
- full ensembles;
- wavelet networks;
- any future dynamic or contact-state formulation.

Training-only losses and optimizers have no direct inference cost, but they still require Python, ONNX, and PLC parity for the selected model.

Stage Routing

The register maps techniques to the remaining roadmap.

Next stage	Frozen work
Stage 2	Gradient diagnostics, normalized weights, ReLoBRaLo, curriculum
Stage 3	PF-A reproduction and corruption stress
Stage 4	Data-only ladder, Fourier features, SIREN, exact periodic construction
Stage 5	Complex harmonic coefficient residuals
Stage 6	Spectral, Sobolev, and conditional wavelet guidance
Stage 7	Mean and centered-shape heads
Stage 8	Weak compliance only if bootstrap support exists
Stage 9	Temporal PF-A plus bounded GRU residual
Stage 10	Sparse and symbolic residual discovery
Stage 11	Uncertainty and anchor-trust calibration
Stage 12	Conditional PCGrad, CAGrad, augmented Lagrangian, and adaptive sampling
Stage 13	Weak-form, certification, synthetic corruption, MMT, and contact oracles
Stage 14	Only isolated candidates that pass matched controls
Stage 15	Official verification and deployment parity

The immediate next implementation remains Stage 2. Physics candidates are not trained before the instrumentation can show whether the objectives cooperate or conflict.

Machine-Readable Evidence

The canonical Stage 1 evidence is stored under:

`output/analysis/wave_5_2r/stage1_extended_scientific_technique_discovery/`

Authored inputs

- `stage1_source_register.yaml` ;
- `stage1_technique_register.yaml` .

Generated outputs:

- `stage1_candidate_register.csv` ;
- `stage1_exit_gate_summary.json` .

The validator enforces source identity, field completeness, search-family coverage, causal-variable allowlists, missing-variable exclusion, and the absence of target-derived runtime quantities. Validation result:

Result field	Value
Status	`WAVE52R_STAGE1_VALIDATION_OK`
Techniques	`30`
Real-data techniques	`24`
Oracle-only techniques	`3`
Excluded techniques	`3`
Search families	`13`

The `24` real-data count contains `13` active and `11` conditional techniques.

Exit-Gate Audit

Gate	Result	Evidence
All required search families covered	pass	`13 / 13`
Every technique has a source	pass	`30 / 30`
Every technique has a local formulation	pass	`30 / 30`
Every technique has a matched control	pass	`30 / 30`
Every technique has a falsification rule	pass	`30 / 30`
Every technique has deployment assessment	pass	`30 / 30`
Real-data roster has no missing variables	pass	`0` violations
Target-derived runtime variables	pass	`0`
Unobservable mechanisms excluded or oracle-only	pass	explicit roster
Candidate register frozen	pass	content-hashed YAML and JSON summary

Decision: freeze the Stage 1 candidate register and authorize Stage 2, Evaluation And Optimization Instrumentation.

This authorization is for non-training Stage 2 implementation and validation. Any later training campaign retains the repository campaign-plan and explicit training-approval gates.

Primary References

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Conclusion

Stage 1 expands the PINN program without weakening scientific discipline.

The most promising direction is not a monolithic full-PINN with every theory inserted as a loss. It is a controlled sequence of hybrids:

- exact periodic representation;
- strong Polynomial-Fourier anchor;
- bounded learned discrepancy;
- direct complex harmonic and derivative supervision;
- explicit mean and shape decomposition;
- weak mechanics only where the data support the prior;
- optimization methods activated only after their failure mode is measured.

  The network can compensate for missing analytical detail. It cannot make unmeasured mechanical states identifiable. The frozen roster preserves both facts and provides a falsifiable path through Stages 2 through 15.